EDS 223: Homework Assignment 2

Exploring Patterns of Environmental Justice: Historical HOLC Redlining and Los Angeles County

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Load packages

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(stars)
library(gt)
```

Read data

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen                    <-                    sf::st_read(here::here("data",
"ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))

# For EJ Screen: filter to a county you are interested in
los_angeles <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))

# Read in Bike and Pedestrian Crash Reports data
crashes <- sf::st_read(here::here("data", "SWITRS_LA","Crashes.csv"))

# Read in HOLC Redlining data
redlining  <-  sf::st_read(here::here("data",  "mapping-inequality",  "mapping-
inequality-los-angeles.json"))
```

Preparing Coordinate Reference System (CRS) for Part 1

```
# Check if the los_angeles CRS and redlining CRS match
if(st_crs(los_angeles) == st_crs(redlining)) {
```

```
    print("CRS matches")
} else {
    warning('CRS does not match')
}
```

```
Warning: CRS does not match
```

```
# Check CRS of los_angeles
st_crs(los_angeles) # the CRS is EPSG:3857
```

```
# Transform CRS of redlining to match CRS of los_angeles
redlining <- st_transform(
    redlining,
    crs = st_crs(los_angeles)
    )
```

```
# Confirm the tranformation worked and the CRS now matches
if(st_crs(los_angeles) == st_crs(redlining)) {
    print("CRS matches")
} else {
    warning('CRS does not match')
}
```

```
[1] "CRS matches"
```

Color palette for map and figures

```
# See the unique values in grade column of redlining
unique(redlining$grade)
```

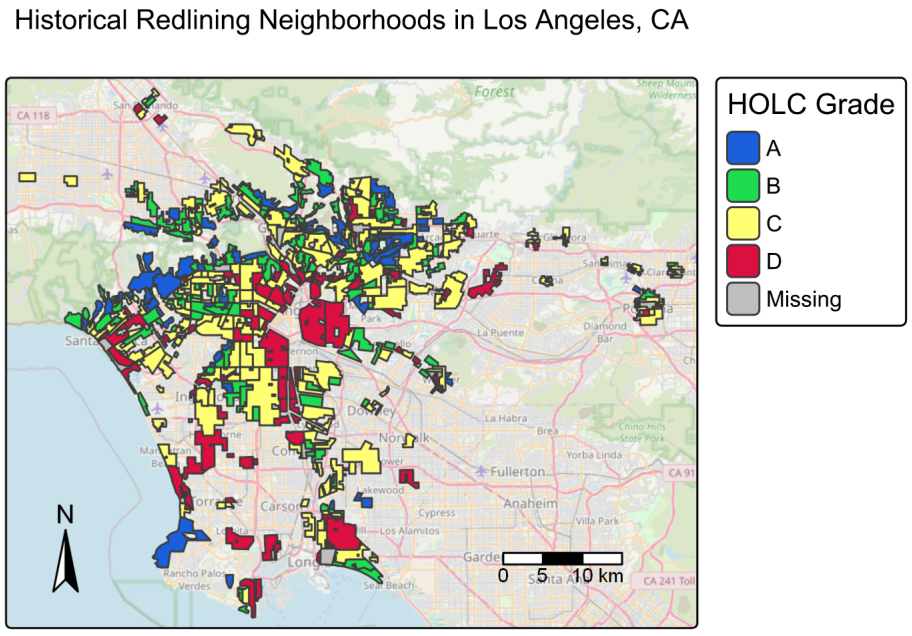
```
[1] "A" "B" "C" "D" NA
```

```
# Create a color palette for the different values in the grade column of redlining
redlining_palette <- c(
      "A" = "#1a63e0",
      "B" = "#20e050",
      "C" = "#ffff77",
      "D" = "#e01044",
      "Missing" = "#c8c1c3")
```

Part 1: Legacy of redlining in current environmental (in)justice

Create a map of historical redlining neighborhoods in LA County

```
# Use tmaps to visualize the neighborhoods in LA County that have recieved an
HOLC grade
tm_shape(redlining) +
    tm_polygons(fill = "grade",
                title = "HOLC Grade",
                palette = redlining_palette) +
    tm_title("Historical Redlining Neighborhoods in Los Angeles, CA") +
    tm_basemap("OpenStreetMap", alpha = 0.6) +
    tm_compass(position = c("left", "bottom")) +
    tm_scalebar(position = c("right", "bottom"), text.size = 0.6, breaks = c(0,
5, 10, 15))
```



Create a table of the HOLC/no-HOLC data in LA County

```
# Reprojecting can create invalid geometries (ex. the polygons have overlapping
vertices)
# st_make_valid() rebuilds these geometries without removing any points, so the
join can run without errors
los_angeles <- st_make_valid(los_angeles)
redlining <- st_make_valid(redlining)
```

```r
# Join los_angeles and redlining
# The CRS for the two datasets have to match because the only thing the two datasets have in common are their CRS
join_los_angeles_redlining <- st_join(los_angeles, redlining)

# Drop geometry column from joined data
# Now that the datasets are joined, we no longer need the geometry column (code will run faster without it)
join_los_angeles_redlining <- st_drop_geometry(join_los_angeles_redlining)

# Create a table summarizing the percentage of census block groups that fall within a HOLC grade and the percentage of census block groups that do not have a HOLC grade
los_angeles_redlining_percent <- join_los_angeles_redlining |>
    mutate(grade = ifelse(is.na(grade), "No HOLC Grade", grade)) |>
    count(grade) |>
    mutate(percent_of_grade = (n / sum(n)) * 100)

# Render the table
gt(los_angeles_redlining_percent) |>
    tab_header(title = 'Percent of Census Block Groups Within Each HOLC Grade in LA County')
```

Percent of Census Block Groups Within Each HOLC Grade in LA County

grade	n	percent_of_grade
A	449	4.99555
B	1239	13.78505
C	3058	34.02314
D	1346	14.97552
No HOLC Grade	2896	32.22074

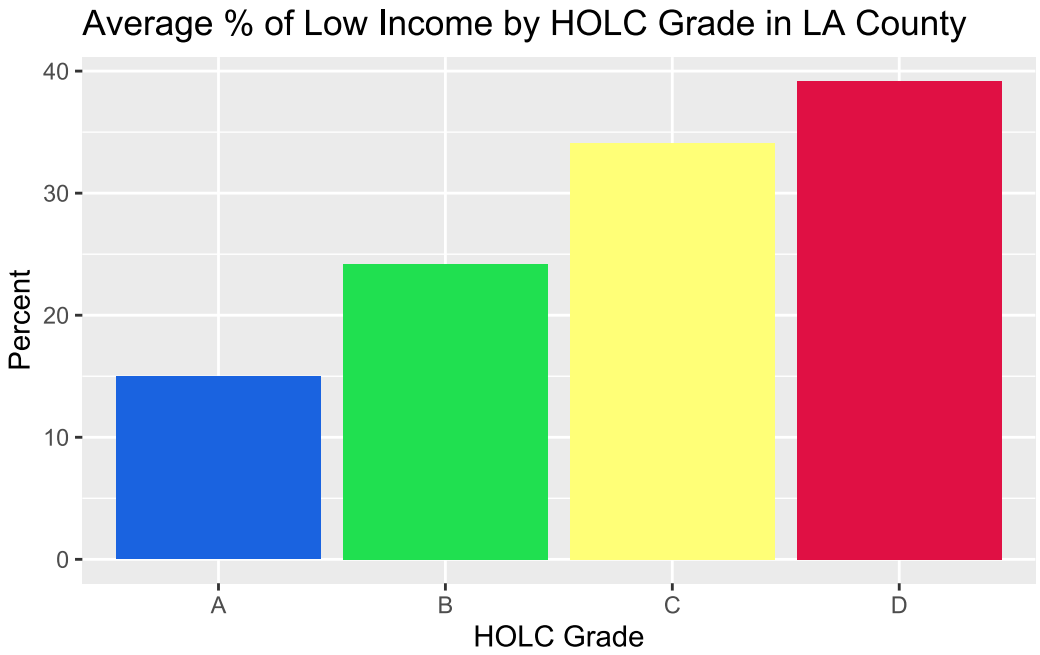
Visualizations summarizing the percentages of Low Income, and Low Life Expectancy and PM2.5 Percentiles within each HOLC grade

Low-Income

```r
# Find the mean for the percentage of low income individuals per HOLC grade in LA County
```

```
mean_low_income_pct <- join_los_angeles_redlining |>
    group_by(grade) |>
    drop_na(grade) |>
    summarise(avg_pct_low_inc = (mean(LOWINCPCT, na.rm = TRUE) * 100)) # create
a column of the averages

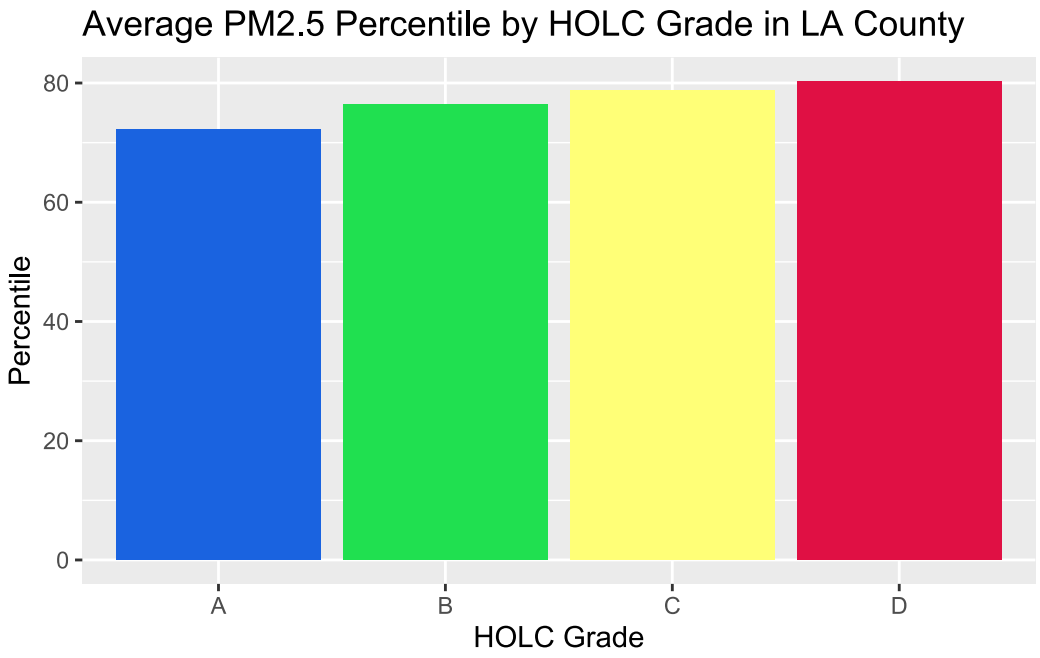
# Create a bar chart to display mean_low_income_pct
ggplot(mean_low_income_pct, mapping = aes(x = grade, y = avg_pct_low_inc, fill
= grade)) +
    geom_col() +
    labs(title = 'Average % of Low Income by HOLC Grade in LA County',
         x = 'HOLC Grade',
         y = 'Percent') +
    scale_fill_manual(values = redlining_palette) +  # set bar colors
    theme(legend.position = "none") # hide legend
```



PM2.5

```
# Find the mean for the PM2.5 percentiles per HOLC grade in LA County
mean_pm25_percentile <- join_los_angeles_redlining |>
    group_by(grade) |>
    drop_na(grade) |>
    summarise(avg_pm25 = mean(P_PM25, na.rm = TRUE))
```

```
# Create a bar chart to display mean_pm25_percentile
ggplot(mean_pm25_percentile, mapping = aes(x = grade, y = avg_pm25, fill =
grade)) +
    geom_col() +
    labs(title = 'Average PM2.5 Percentile by HOLC Grade in LA County',
         x = 'HOLC Grade',
         y = 'Percentile') +
    scale_fill_manual(values = redlining_palette) + # set bar colors
    theme(legend.position = "none") # hide legend
```

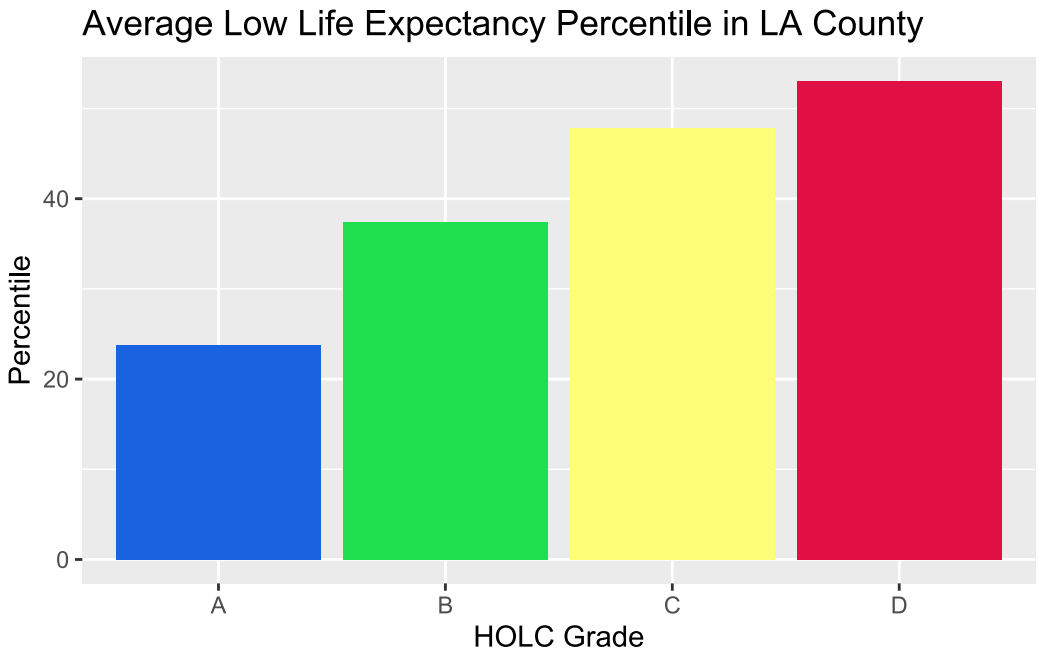


Low Life Expectancy

```
# Find the mean for the low life expectancy percentiles per HOLC grade in
LA County
mean_life_expt_percentile <- join_los_angeles_redlining |>
    group_by(grade) |>
    drop_na(grade) |>
    summarise(avg_life_expt = mean(P_LIFEEXPPCT, na.rm = TRUE))

# Create a bar chart to display mean_life_expt_percentile
ggplot(mean_life_expt_percentile, mapping = aes(x = grade, y = avg_life_expt,
fill = grade)) +
    geom_col() +
    labs(title = 'Average Low Life Expectancy Percentile in LA County',
```

```
        x = 'HOLC Grade',
        y = 'Percentile' ) +
    scale_fill_manual(values = redlining_palette) + # set bar colors
    theme(legend.position = "none") # hide legend
```



Interpretation of the Results

In all three figures, conditions generally get worse from Grade A to Grade D neighborhoods. The average share of low-income residents goes from about 15% in Grade A neighborhoods to 39% in Grade D neighborhoods. The average percentile for low life expectancy also increases from about 24 to 53, and PM2.5 percentiles go from about 72 to 80. The difference in air pollution is smaller, but it still follows the same pattern. Overall, these results show that neighborhoods that were graded lower during redlining tend to have worse economic and health outcomes today.

Residents in lower-graded neighborhoods were often denied home loans, likely making it harder to buy homes and build up wealth. The lower-graded neighborhoods likely had lower property values and less political influence, which may have made them more likely to have industrial areas or freeways nearby. Both can contribute to higher air pollution, and long-term exposure to pollution (along with economic disadvantages) can affect people's health. This could help explain why Grade D neighborhoods also have worse life expectancy outcomes. While this analysis does not prove that redlining directly caused these differences, it shows a clear relationship between historical redlining grades and conditions today.

Part 2: Legacy of redlining in pedestrian and bike accidents

Preparing NA Values for Part 2

```
# Determining if LATITUDE and LONGITUDE or POINT_X and POINT_Y have less NAs
sum(crashes$LATITUDE == "NA") # has 97,350 NAs
```

```
[1] 97350
```

```
sum(crashes$POINT_X == "NA") # has 9,484 NAs
```

```
[1] 9484
```

```
# POINT_X and POINT_Y should be used because there are much less NAs than LATITUDE
and LONGITUDE
# note: if POINT_X has 9,484 NAs, so does POINT_Y and same for LAT/LONG which
is why we only have to sum the NAs in one of each

# Converting "NA" strings to NA values
crashes$POINT_X[crashes$POINT_X == "NA"] <- NA
crashes$POINT_Y[crashes$POINT_Y == "NA"] <- NA
# As a text string, R does not think of NA as a missing value, it sees it as
two letters
# For this analysis, we need R to recognize NA values not "NA" strings
```

Matching the CRS and turning a df into an sf object for Part 2

```
st_crs(redlining) # uses WGS84 and CRS is ESPG:3857
# In the metadata for SWITRS, it says that POINT_X and POINT_Y use WGS84
```

```
# We loaded in crashes as a dataframe, it needs to be an sf object (dataframes
cannot recognize geographic coordinates and we cannot have POINT_X/Y treated
like numbers)
crashes <- crashes |>
     select( # pull out important columns (no LAT/LONG because we are using
POINT_X/Y instead)
        ACCIDENT_YEAR,
        PEDESTRIAN_ACCIDENT,
        BICYCLE_ACCIDENT,
        COUNT_PED_KILLED,
        COUNT_BICYCLIST_KILLED,
```

```
        POINT_X,
        POINT_Y
        ) |>
    # keep row if the accident year is between 2020 and 2025
    filter(
        ACCIDENT_YEAR >= as.integer("2020") &
            ACCIDENT_YEAR <= as.integer("2025")
        ) |>
    # keep row if there was a pedestrian or bicyclist accident
    filter(
        PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y"
        ) |>
    # missing values in coordinates is not allowed so drop NAs
    drop_na(POINT_X, POINT_Y) |>
    # change crashes from a dataframe to an sf object
    st_as_sf(coords = c("POINT_X", "POINT_Y"), crs = "EPSG:4326") |>
    st_transform(st_crs(redlining)) # match the CRS to the CRS of redlining
```

```
# Confirm we have matched the CRS
if(st_crs(redlining) == st_crs(crashes)) {
    print("CRS matches")
} else {
    warning('CRS does not match')
}
```

```
[1] "CRS matches"
```

Create a table and figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade

```
# Join crashes dataframe with redlining dataframe
join_crashes_redlining <- st_join(crashes, redlining)

# Create a new data frame using the joined dataframe that summarized the crashes
and fatalities of pedestrians and bicyclists
crashes_redlining <- join_crashes_redlining |>
    st_drop_geometry() |>
    filter(!is.na(grade)) |> # drop NAs in grade column
    group_by(grade) |>
    summarize(
        total = sum(
                    PEDESTRIAN_ACCIDENT == "Y", # count pedestrian accidents
```

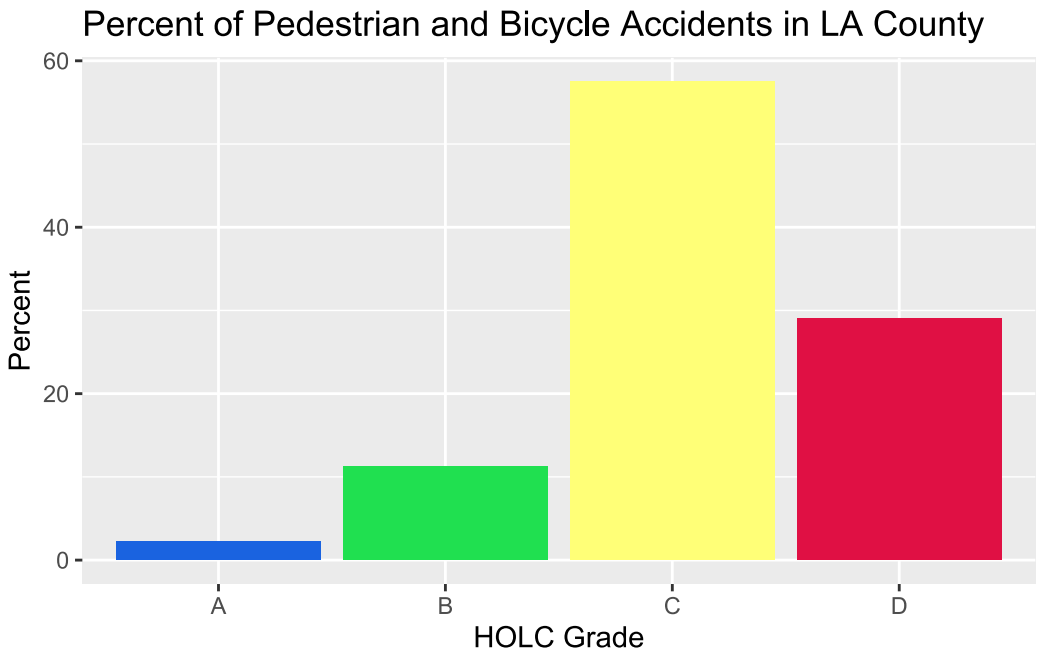
```
                    BICYCLE_ACCIDENT == "Y", # or bicycle accidents
                    as.integer(COUNT_PED_KILLED) != 0, # count people killed in
pedestrian accidents
                as.integer(COUNT_BICYCLIST_KILLED) != 0, # and bicyclist accidents
                    na.rm = TRUE) # ignore missing values instead of returning NA
    ) |>
    mutate(percentage = total/sum(total) * 100) # adding percentage column

# Render table of crashes_redlining
gt(crashes_redlining) |>
  tab_header(title = 'Percent of Pedestrian and Bicycle Accidents Within Each
HOLC Grade in LA County')
```

Percent of Pedestrian and Bicycle Accidents Within Each HOLC Grade in LA County

grade	total	percentage
A	367	2.222761
B	1859	11.259161
C	9491	57.482890
D	4794	29.035189

```
# Bar plot of percentages of the sum of bike and pedestrian crashes and fatalities
ggplot(crashes_redlining, mapping = aes(x = grade, y = percentage, fill = grade))
+
    geom_col() +
    labs(
        x = 'HOLC Grade',
        y = 'Percent',
        title = 'Percent of Pedestrian and Bicycle Accidents in LA County',
        fill = 'HOLC Grade'
    ) +
    scale_fill_manual(values = redlining_palette) + # set bar colors
    theme(legend.position = "none") # hide legend
```



Interpretation of Results

Pedestrian and bicycle crashes and fatalities in HOLC-graded areas of LA County were heavily concentrated in lower-graded neighborhoods. Grade C neighborhoods accounted for 57.5% and grade D for 29.0%, while Grade B accounted for 11.3% and Grade A for only 2.2%. Together, the two lowest grades (C and D) account for about 86.5% of the total. However, Grade C neighborhoods cover much more of Los Angeles than Grade A neighborhoods, so some of this difference may just reflect their larger size rather than more dangerous streets.